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A new model for small target adult image recognition

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Abstract

With the advent of the era of 5G high speed and massive information, the audience of netizens is developing at a younger age, and the proliferation of adult images on the Internet is becoming more and more serious. The recognition and filtering of adult pictures has become very important to prevent young people from contacting harmful information. But, because of the wide variety of adult images, automatic recognition in these pictures is a very difficult task. This paper proposes a new model based on Residual neutral network. Our method consists of two parts. The first part is to divide images into "large target skin color image" and "small target skin color image" according to the proportion of skin color pixels of the image; The second part is to input the two types of images into two different models for recognition. According to the results of comparative experiments, the proposed network can further improve the classification accuracy of adult images of small targets with excessive background noise, which is better than other single model.

Keywords: Deep-learning; Image recognition,; Adult images; Residual neutral network

1. Introduction

Image recognition is one of the important technologies for digital image processing in the information age. Its appearance makes it possible for people to process massive amounts of physical information under the current situation of explosive growth of data. The process of traditional image recognition technology is mainly divided into information acquisition, information preprocessing, feature extraction and selection, and classifier design and

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decision-making. Thanks to the rapid development of deep learning[5], image recognition technology has a new development direction and research route.

In recent years, image recognition algorithms based on deep learning have developed rapidly, and compared with traditional image recognition algorithms, image recognition algorithms based on deep learning, the depth of the algorithm network determines whether the algorithm can extract deeper features in the learning image, and networks with relatively deep layers can extract deeper features. And the accuracy of image recognition algorithms based on deep learning is generally improved by nearly 20% to 30%, so it is very necessary for the research of image recognition technology based on deep learning.

2. Adult Image Recognition Methods

2.1. Skin-Color Method

The adult image recognition method based on skin-color can be roughly divided into two steps: The first step is to research on the skin-color with the original image and establish a skin-color model to determine whether each pixel is a skin-color; The second step is to use the pixel information about skin-color from the first step and determine the proportion threshold of the skin-color pixel based on experience. Based on the above information, calculate the global image feature map, and then train a classifier, such as support vector machines (SVM) , multilayer perceptron (MLP) , or decision trees. In most cases, researchers will use statistical values as features, such as the proportion of skin-color pixels in an image. To study the skin color model, we must first define the skin color, and there are three ways to define skin color:

- By defining the skin color interval and constraints manually;
- By calculating the color histogram;
- By parametric color distribution functions;

After defining the skin color model, each pixel of the image can be divided into skin color pixels or non-skin color pixels, and the following recognition system has been established based on this information.

All related recognition methods are based on the same procedure, apply color models to find skin color pixels, and then calculate global image feature map based on this information and use them to train classifiers, such as support vector machines (SVM) , multi-layer perceptron (MLP) or decision trees, and in most cases, the feature uses a statistical value similar to the proportion of skin color pixels. If the above methods are classified in detail, they can be divided into two categories:

- Color feature map.
- Combined color feature map and face detection.

The main advantage of a model based on color classification is that it can calculate features quickly and is easy to implement. But the shortcomings are also obvious. Because sand, wood, etc. are similar in color to the skin, they are easily marked as skin color pixels by the color model, and the color distribution of some non-adult images is similar to adult images, such as people wearing swimwear.

2.2. Shape Information Method

In the field of adult image classification, the method of feature extraction based on shape information is also very popular. However, this method has many limitations. Many different kinds of things have similar shapes, which introduces a high level of abstraction. In order to implement training standard machine learning algorithms, this highly abstract process filters out most of the poses. It also means that too much information is lost, and the recognition methods based on shape information implicitly include the process of judging whether the pixel is skin color, which makes the performance of the overall algorithm still affected by the skin color model.

2.3. Local Features Method

Extracting local features from the original image is a very popular method of target detection and recognition. Feature vectors are usually used to describe image regions. One of the most famous is SIFT, and it is usually used in combination with DOG key points detector.

Adult image recognition and detection based on local features is a very popular method, but compared with methods based on skin color or shape information, it is a very small part. Due to the different data sets used, the controlled variable method cannot be fairly compared with other methods. However, according to the experimental results given by the researchers, the method based on local features is more effective, so this is also an important research direction.

2.4. Deep Learning Method

With the development of artificial intelligence (AI) and neural networks[9], deep neural networks have achieved remarkable results in the fields of image recognition and target detection. Image recognition has a new development direction and research space. Various networks emerge in endlessly, such as AlexNet, VGGNet, ResNet[3, 4, 5], DenseNet[6]. And new ideas based on improving these models, such as reducing the amount of calculation and accelerating convergence[1, 2]. Some researchers started with key visual cues and proposed an adaptive learning system based on convolutional neural networks[7, 8, 10, 12]. By training the convolutional neural network to coarsely detect the image skin and facial regions, and then further finely detect more skin regions, the accuracy of the system method is significantly better than other traditional manual feature extraction method.

3. Propose a new model

3.1. Basic Content

- **Residual neural network (ResNet):** According to the Universal Approximation Theorem (Universal Approximation Theorem), as long as there is enough capacity, we can only use a single-layer feedforward neural network to represent any function. In this way, the network layer will be very huge, and therefore it is prone to problems such as over-fitting. In order to solve the above problems, our common practice is to deepen the network layer and reduce the size of each network layer. However, increasing the network depth by simply superimposing the network layer has caused problems such as the disappearance of the gradient. As the network depth continues to increase, its performance will gradually become saturated or even begin to decline. In order to solve this problem, Kaiming He et al. proposed the ResNet[5] network structure, which greatly solves the problems of gradient disappearance and gradient explosion caused by too deep network layers. The core idea of the ResNet network is to introduce a "shortcut connection" that can skip one or more layers. For example, see Fig. 1. ResNet enables the original input information to be transmitted to the following network layer through the "highway", thus, the amount of computation required for learning each layer of neural network is greatly reduced, and only the residual output from the previous network layer needs to be learned instead of the whole input.

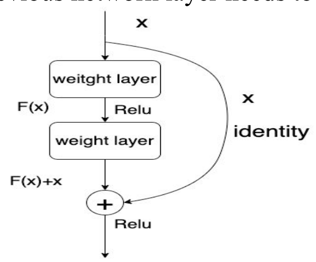

Fig. 1. "shortcut connection"

3.2. A New Model Using Rough Classification

We propose a new model that can improve the accuracy of adult image recognition for small targets without reducing the accuracy of ordinary adult image recognition. The structure of model is shown in Fig. 2. The images first enter a filter to determine whether the skin color ratio in the image reaches the threshold, that is, whether the image is likely to become a small target adult picture. If it exceeds the threshold, enter the B model for recognition, if not, enter the A model, and finally get the image category. The skin-color model uses the skin color definition based on the color histogram researched by Yang et al. to discriminate skin color pixels in the image. Model a and model b are trained based on the residual neural network 101.

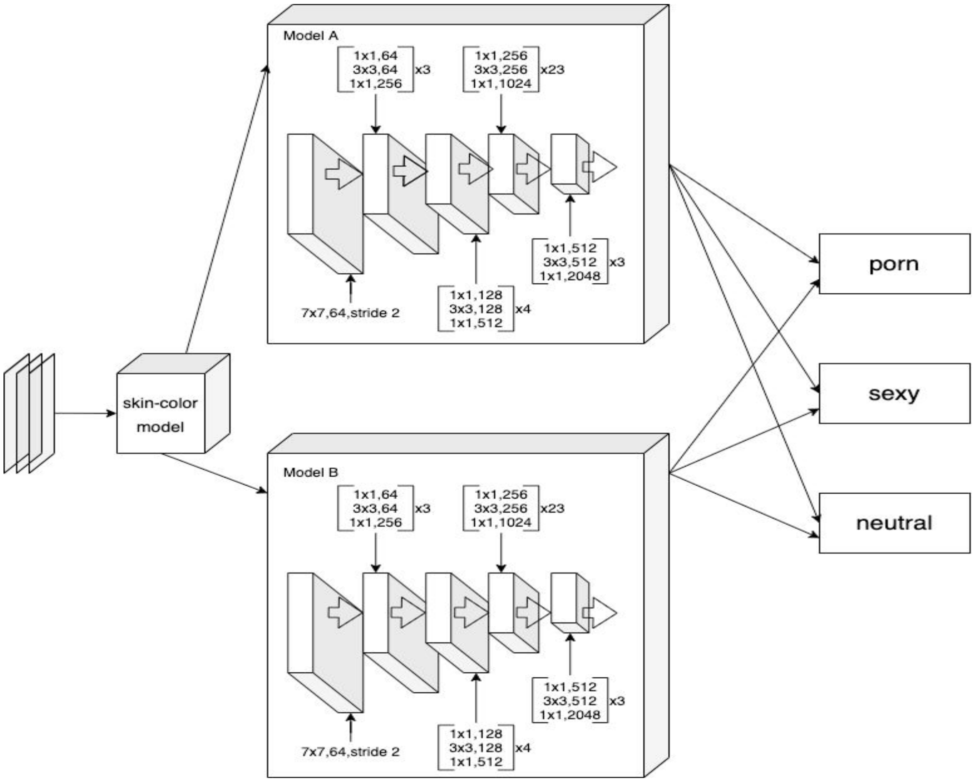

Fig. 2. structure of model

3.3. Dataset

Due to the particularity of the field of adult image recognition, the dataset in this article is downloaded from the Internet by a crawler program. There are about 4W images, divided into three categories, namely porn, sexy, and neutral. Each category of pictures is about 1.4W. And use data enhancement technology to construct about 1W small target adult images, which are used to train a model that recognizes small target adult images. As shown in Fig. 3.

(a) (b)

Fig. 3. (a) is a small target adult image;(b) is an ordinary adult image

3.4. Results and analysis

We select the accuracy rate as the evaluation index, the calculation method is as formula (1). Among them, TP represents the number of adult images correctly identified as adult images by the model, TN represents the number of normal images correctly identified as normal images by the model, and P and N represent the number of adult images and non-adul images, respectively. The confusion matrix of the model test accuracy is shown in Fig. 4.

$$Acc = \frac{TP + FP}{T + F} \tag{1}$$

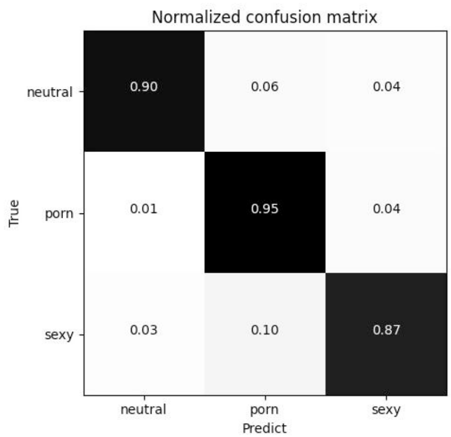

Fig. 4. The confusion matrix

In order to comprehensively evaluate the performance of the method in this paper, the accuracy of the method in this paper is compared with other methods. The results are shown in Table 1.

Table 1.Acc of different models.

Model	Acc(%)
AlexNet	92.1
ResNet	93.3
New Model	95.0

It can be seen from the Table 1 that the effect of the models based on the convolutional neural network is better when facing ordinary adult pictures. However, because the test set contains small-target adult images, their effects are still far from the new model proposed in this paper, and they are not stable enough.

Operating environment is as follows:

- CPU: Intel (R) Core (TM) i7-9700K CPU;
- GPU: NVIDIA RTX 2080Ti;
- Memory: 32GB;
- OS: Linux.

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